

THERMACART · CALCULATION

ThermaCart sizing calculations

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-02	Amish Chadha	O3 decision of 2026-10-02 recorded (C5 freezer charging with a required conditioning step); no figure changed
0.3	2026-10-01	Amish Chadha	Constructable design (TCT-DDR-003) applied to mass, bail thermal break and cost; budget treated as a value-engineering target, R16 reported against it
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); dark finish as the design case, paint mass and cost added, R1, R5 and R14 restated, results table updated
0.1	2026-09-25	Amish Chadha	First issue for TRL 3 (geometry and mass, storage per grade, pressure and wall stress, 2 mm wall option, hold times, recharge, handle temperature, keying, cost)

ThermaCart sizing calculations

On paper the TC-L cartridge meets twelve of its eighteen requirements, has one at risk, misses one, is over its cost target on one and leaves three that only tests can settle. Version 0.3 follows the constructable design of TCT-DDR-003 (sealed radial screws, bolted lug angles on phenolic washers, a G 1/2 fill port, fixings for every part): the empty mass is unchanged at 1.80 kg, the bail runs cooler on its new thermal break, and the parts cost rises from \$76 to \$80 per cartridge. The budget in `project.yaml` is now treated as a value-engineering target, not a limit (Amish, 2026-10-01). This version applies the recommendations Amish accepted on 2026-09-25 (TCT-DDR-002): the shell carries a matte black finish (O8), so the dark-finish results are now the design values; the C25 band of R1 is 22 to 27 °C (O9); R5 is 38 % for the stock-tube build (O4); and R14 includes a labelled handling rule for H70 cartridges from an oven (O7). The miss is the cold-chain hold time (R10: 2.4 h against 8 h, because the cartridge cannot pull heat out of the cooler air fast enough, although it stores enough energy for 9.7 h) and, reported against the value-engineering targets rather than as a miss, the parts cost (R16: \$80 per cartridge, \$30 over its \$50 target; one cartridge and one frame \$97, \$17 over the \$80 value-engineering target). The one at risk is recharge on a 60 W pad (R12). With the verified Rubitherm datasheet values, the C5 grade now meets R3 (65 Wh), which the TRL 2 estimate did not; C25 meets it by a thin margin. The 2 mm wall evaluated under decision D5 is rejected: it adds only 3.5 % volume and its walls would yield under the vacuum that forms when a cartridge is frozen. Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [C3], is the line of that script's output that carries it.

Safety: These are first-principles estimates for a paper proof of concept. They do not show that a cartridge is leak-tight, drop-safe or fit for food contact, and they are not a substitute for tests. ThermaCart is not qualified for vaccines or medicines. See TCT-PRC-001, Safety.

■ Scope and method

The note checks every requirement in TCT-REQ-001 v0.5 against the design in TCT-PRC-001 v0.5, the constructable design of TCT-DDR-003 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS`, its derived dimensions and its build123d solids, so the inner volume, part volumes, areas and key positions used here are those in the STEP files and in drawing TCT-DWG-001. It also reads `bom/bom.csv` and the value-engineering target `budget_usd` in `project.yaml`. Run it from the repo root with `python docs/04-calcs/sizing.py`.

■ Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
PCM data	RT 5 HC: 250 kJ/kg from -2 to 13 °C, melt 5 to 6 °C, 0.85 kg/L solid, 0.76 kg/L liquid at 20 °C, maximum operating temperature 45 °C. RT 25 HC: 230 kJ/kg from 16 to 31 °C, melt 22 to 26 °C, 0.88 and 0.77 kg/L (liquid at 40 °C), maximum 65 °C. RT 70 HC: 260 kJ/kg from 62 to 77 °C, melt 69 to 71 °C, about 0.9 and 0.8 kg/L (liquid at 80 °C), maximum 90 °C. All: cp 2 kJ/(kg·K), k 0.2 W/(m·K), capacity $\pm 7.5\%$	Rubitherm datasheets for RT 5 HC , RT 25 HC and RT 70 HC , read 2026-09-25
Usable storage	Latent heat = capacity at the low side of the tolerance minus 2 kJ/(kg·K) x 15 K sensible; usable = latent plus sensible over the ± 3 K band of R3; latent spread evenly over the melt range	Conservative reading of the datasheets
Fill	90 % of the inner volume, filled as liquid at the grade's maximum operating temperature (45, 65 and 90 °C), then sealed with air at 1 atm; liquid expansion 0.001 /K, solid 0.0003 /K	Expansion coefficients are typical paraffin values, to confirm
Aluminium	6063-T52 tube, minimum yield 110 MPa, design factor 1.5 (allowable 73 MPa); E 69 GPa; 2.70 kg/L, 0.9 kJ/(kg·K)	ASTM B221 minimum
Air side	Natural convection $h = 1.32 (\Delta T/0.15 \text{ m})^{0.25} \text{ W}/(\text{m}^2\cdot\text{K})$ plus radiation; emissivity 0.90 for the decided matte black finish (TCT-DDR-002, O8), 0.10 for mill-finish aluminium as a comparison; 85 % of the outer area exposed	Handbook correlations
Cold box (R10)	25 L cooler, 0.5 W/K, 32 °C ambient; 5 kg of produce precooled to 5 °C (3.9 kJ/(kg·K)) plus 2 kJ/K of air and liner; cartridges conditioned to 2 °C; contents and cooler air lumped, no direct contact between produce and cartridge	As in TCT-REQ-001; contact would help and is not credited
Hot box (R11)	GN carrier, 0.25 W/K, 25 °C ambient; 4 kg of food at 75 °C (3.5 kJ/(kg·K)) plus 2 kJ/K of liner; H70 charged to 85 °C	As in TCT-REQ-001
Charging (R12)	Still-air freezer at -18 °C, refrigerator at 2 or 4 °C, fan oven at 85 °C ($h 15 \text{ W}/(\text{m}^2\cdot\text{K})$ plus radiation), 60 W pad with 80 % of its output entering the cartridge	Screening values
Burn thresholds (R14)	10 s contact: about 55 °C for bare metal and about 70 °C for plastics and rubber	Commonly quoted from ISO 13732-1; not checked against the standard (paywalled)
Cost	Indicative one-off USD prices in <code>bom/bom.csv</code> , not quotes; PCM at \$10/kg; <code>budget_usd</code> read as a value-engineering target	To be replaced with quotes before any build

■ A. Geometry and mass (R2, R4, R5)

- **Envelope.** With the bail stowed the cartridge is 323.4 x 152.4 x 63.5 mm, inside the GN 1/3 slot (325 x 176 mm, 65 mm deep) with margins of 1.6, 23.6 and 1.5 mm [A1]. The TRL 2 fixed handle left only about 14 mm between grip and end cap, too little for fingers; the folding bail stows inside the envelope and gives a 41 mm finger gap when raised [A3]. The bail now pivots on two bolted lug angles (TCT-DDR-003, P5).
- **Inner volume.** 146.05 x 44.45 x 254.9 mm, or 1.655 L [A2], slightly more than the 1.62 L of TRL 2, because a thinner 3.175 mm (1/8 in) flange let the tube grow from 270 to 280 mm.
- **Mass.** The model's aluminium parts come to 1.657 kg, and seals, screws, bolts, pins, the anodised aluminium fill plug, grip, label, indicator, adhesive and about 15 g of primer and paint add 0.140 kg, so the empty cartridge is 1.80 kg [A4, A5] against about 1.5 kg at TRL 2. The caps are the reason: a garage-built cap is a solid stack of flange, gland spacer and plug (about 100 cm³ each) rather than the milled, pocketed cap assumed at TRL 2.

- **Filled.** C5 2.90 kg, C25 2.92 kg and H70 2.98 kg, with 1.104, 1.119 and 1.180 kg of PCM [A6]. R4 (3.0 kg) is met with a thin margin. The PCM fraction is 38.1 to 39.6 %, which meets the 38 % target set for the stock-tube build by TCT-DDR-002 (O4) with little margin; the 50 % target stays for a later extrusion.

■ B. Usable storage per grade (R1, R3)

R1 is met for all three grades. RT 25 HC melts over 22 to 26 °C, which lies inside the C25 band of 22 to 27 °C set by TCT-DDR-002 (O9); the earlier 24 to 27 °C band would not have been met.

Table 2. Storage within ± 3 K of the melt point [B1].

Grade	PCM	Latent (low side)	Usable in band	Per cartridge	At nominal datasheet value
C5	1.104 kg	201.2 kJ/kg	213.2 kJ/kg	65.4 Wh	71.2 Wh
C25	1.119 kg	182.8 kJ/kg	194.8 kJ/kg	60.5 Wh	65.9 Wh
H70	1.180 kg	210.5 kJ/kg	222.5 kJ/kg	72.9 Wh	79.3 Wh

- All three grades meet R3 (60 Wh). The TRL 2 figures (55, 61 and 67 Wh) used assumed usable heats of 180, 200 and 220 kJ/kg; the datasheets support more for C5 and H70 and less for C25.
- Energy density on filled mass is 22.6, 20.8 and 24.5 Wh/kg [B2].
- **2 mm wall (D5).** A metric 150 x 50 x 2 mm tube of the same length holds 1.712 L, only 3.5 % more, and would raise C5 to 67.7 Wh [B3]. Section C shows its walls would yield. It is not needed for R3 and is not adopted.

■ C. Pressure and wall stress (R8, D5)

Filling as liquid at the grade's maximum operating temperature fixes the gauge pressure at zero at that temperature. Everything colder makes a partial vacuum, because the PCM shrinks by about 12 % on freezing and the trapped air cools.

Table 3. Gauge pressure with 10 % ullage [C1].

Grade	Sealed at	Solid at -25 °C	20 K above the limit
C5	45 °C	-0.63 bar (ullage 20.8 %)	+0.29 bar at 65 °C
C25	65 °C	-0.68 bar (ullage 22.1 %)	+0.29 bar at 85 °C
H70	90 °C	-0.64 bar (ullage 18.5 %)	+0.29 bar at 110 °C

- **R8 is met:** 0.00 bar at the fill temperature by construction and +0.29 bar even 20 K beyond the datasheet limit, against 0.5 bar [C2].
- **Walls.** Treating the tube as a closed frame loaded on all four walls, the corner bending stress is 56 MPa at -0.68 bar, 83 MPa at full vacuum and 24 MPa at +0.29 bar, against an allowable of 73 MPa [C3]. The 3.175 mm wall carries the worst service vacuum with margin. Full vacuum could only arise in a cartridge sealed with no air; it would exceed the allowable stress but not the yield. The broad faces deflect about 0.8 mm inward at -0.68 bar [C5].
- **2 mm wall (D5).** The same vacuum gives 141 MPa, 1.28 times the minimum yield; its allowable pressure is only 0.35 bar [C4]. Rejected.

- **Caps.** At +0.29 bar each cap carries 190 N, 32 N per radial M5 screw [C6]. Under vacuum the flange bears on the tube end.
- **Drop (R6, indicative).** A molten cartridge dropped 1 m end-on and stopping in 4 mm decelerates at about 250 g; if the gas ullage did not cushion the liquid, the surge could reach 5.0 bar on the lower cap, 541 N per screw [C7]. The screws have ample capacity (M5 A2-70 proof load about 6,400 N), but that surge is ten times the service pressure on the walls and the O-ring. R6 needs a drop test at TRL 4.

■ D. Hold times (R10, R11)

The outer area is 0.1138 m² of tube, 0.0503 m² of fins and 0.0155 m² of end faces, 0.1526 m² effective [D1].

R10 is not met. Two C5 cartridges store 131 Wh within the band, enough for 9.7 h at the 13.5 W leak [D2], so energy is not the limit. The limit is heat transfer from the cooler air into the cartridges. With the decided matte black finish (emissivity 0.9), still air and radiation give about 7.3 W/(m²·K), 1.11 W/K per cartridge, so the contents settle near 10 °C while the PCM melts and leave the 2 to 8 °C band after 2.4 h at 32 °C [D3] (4.0 h at 25 °C [D4]); the band holds indefinitely while the PCM melts only below about 19 °C ambient. A bare mill finish would give 1.9 h [D3], and a third dark cartridge 3.4 h [D5]. Under TCT-DDR-002 (O5) the user guidance is to lay produce directly on the cartridges, which is how ice packs are used; the model does not credit that contact, so this is a pessimistic bound, and R10 keeps its 8 h target until a TRL 4 measurement (on hold) shows the contact effect.

R11 is met. One H70 cartridge charged to 85 °C keeps 4 kg of food from 75 °C at 63 °C or more for 13.6 h with the decided dark finish and 7.2 h with a bare finish [D6]. The food alone would last 4.9 h [D7], so the cartridge adds at least 2.3 h. The hot case works because the H70 cartridge runs about 7 K above the food and radiation is stronger at 70 °C.

■ E. Recharge (R12)

A quasi-steady Stefan model adds the resistance of the growing layer of changed PCM (k 0.2 W/(m·K), up to 22 mm from each broad face) to the air-side film.

Table 4. Recharge time [E1, E2, E3].

Case	Bare finish	Dark finish	R12 (8 h)
C5 from 8 °C in a -18 °C freezer	5.6 h	4.2 h	Met
C5 in a 2 °C refrigerator	51.1 h	31.4 h	Not practical
C25 from 28 °C in a -18 °C freezer	2.8 h	2.1 h	Met
C25 in a 4 °C refrigerator	6.6 h	4.6 h	Met
H70 from 25 °C in a fan oven at 85 °C	6.1 h	5.3 h	Met
H70 on a 60 W pad	2.7 h if the heat reaches the PCM; up to 10.3 h if the melt must conduct upward from a 90 °C base		At risk

- The dark finish (O8) is the design case: C5 in a freezer 4.2 h, C25 in a refrigerator 4.6 h and H70 in an oven 5.3 h all meet the 8 h target [E1, E2].
- Charging C5 in a refrigerator, one option in the TRL 2 open questions, would take one to two days, so freezer charging with a conditioning step is the practical route. Amish decided on 2026-10-02 (O3) to charge C5 in a domestic freezer followed by the conditioning step, with the conditioning time printed on the C5 label as a required step.
- The TRL 2 figure of about 2.5 h on a 60 W pad assumed the heat is absorbed as fast as it is supplied. It needs 130 Wh [E3]; paraffin conducts poorly, and the pad must never push the base above the RT 70 HC limit of 90 °C, so the true time lies between 2.7 and 10.3 h and depends on convection in the melt. This is a question for a TRL 4 test.
- **R12 target correction.** The TRL 2 target allowed an oven "at 100 °C or less", which is above the 90 °C maximum operating temperature of RT 70 HC. TCT-REQ-001 v0.3 and later say 85 °C, never above 90 °C.
- The energy flow of one pad charge is 162 Wh in, 130 Wh stored between 25 and 80 °C, 88 Wh released above 63 °C and 42 Wh below it, with 32 Wh lost at the pad [E4] (Figure 2 of TCT-PRC-001).

■ F. Handle temperature (R14)

- **Thermal break.** In the constructable design (TCT-DDR-003, P5) each lug angle stands on two 3 mm phenolic washers and each of its two M4 bolts has a 3 mm phenolic washer under its head; the bolt passes the angle with an air gap, so a steel bolt never carries heat straight into the bail. The eight washers conduct 0.032 W/K and the bail (angles, arms and the bare part of the rod, 0.0117 m² from the model) loses 0.094 W/K to the air, so a bail on a cartridge at 71 or 80 °C settles at 36.8 or 39.1 °C [F1], below the assumed 55 °C bare-metal threshold [F3]. Version 0.2 gave 39.6 and 42.5 °C for the concept bail.
- **Straight from the oven** the bail is at oven temperature, 85 °C. With a 9 min time constant it falls below 55 °C after about 10 min [F2]; the label's 12 min rule keeps its margin. The silicone grip is also at 85 °C at first, above the assumed 70 °C threshold for rubber.
- R14 is **met with the handling rule** decided in TCT-DDR-002 (O7): pad charging is preferred and keeps the bail near 39 °C, and the label tells users to take an H70 cartridge from an oven with oven gloves or to wait 12 min. Without the rule, a cartridge lifted straight out of an oven would not meet the burn threshold. The bail is bare

aluminium on its thermal break and is not painted, so the dark finish does not change these figures. C5 and C25 never exceed 65 °C and meet R14.

■ G. Grade keying (R9)

Key tabs 30 mm wide sit at -43, 0, +43 and -61 mm (C5, C25, H70, W0), and each frame slot is 3 mm wider than its key on each side. Every pairing was checked: each frame accepts only its own grade [G1]. The W0 key overlaps the C5 slot but cannot pass it.

■ H. Cost against the value-engineering targets (R16)

`budget_usd` is a hypothetical value-engineering target, not a spending limit (Amish, 2026-10-01: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens"). Value-engineering target: USD 80 for one C5 cartridge and one frame. Estimated cost of the constructable design: USD 97 (USD 17 over the target).

- One cartridge costs \$80.00, \$30 over its \$50 target, and one frame \$17.00, \$3 under its \$20 target [H1]. The dark finish (O8) adds about \$5 for primer and paint; the parts added for construction (TCT-DDR-003: bonded seals, bolts, phenolic washers, rivets) add \$5, less \$1 for the smaller fill plug. The TRL 2 estimate of \$49 left out the fin bar and finish, priced the caps as simple plates and used a lower tube price; all figures remain indicative, not quoted.
- One C5 cartridge and one frame cost \$97.00, \$17.00 over the \$80 value-engineering target [H1]; supplier quotes (TCT-DDR-002, O6) wait with TRL 4. A set of one cartridge per grade and one frame would be about \$258 [H2]. PCM at \$20/kg instead of \$10/kg would add \$11 per cartridge [H3].
- The build uses plates cut and filed and bars bonded with epoxy, with no milling; the frame needs a hand brake, available at a makerspace.

■ I. Results against every requirement

Table 5. Requirement status from this note [S1 and sections A to H].

ID	Requirement	Value	Target	Status
R10	Cold-chain hold time	2.4 h with the dark finish (1.9 h bare) at 32 °C; energy alone 9.7 h; produce contact not credited [D2, D3]	8 h at 2 to 8 °C	Not met
R16	Low cost, garage-buildable	\$80 per cartridge, \$17 per frame; no milling [H1]	\$50 and \$20 value-engineering targets	Over the value-engineering target by \$30 per cartridge; frame \$3 under
R12	Recharge with common equipment	Dark finish: C5 freezer 4.2 h, C25 refrigerator 4.6 h, H70 oven 5.3 h; H70 pad 2.7 to 10.3 h [E1 to E3]	8 h or less	At risk (pad)
R6	Sealed for life	Seal design reviewed; drop surge up to 5.0 bar flagged [C7]	No leak over 1,000 cycles and a 1 m drop	Not verifiable at TRL 3
R7	Long life	Paraffins plausible; supplier cycling data not yet checked	1,000 cycles, under 10 % loss	Not verifiable at TRL 3
R17	Open and documented	Envelope, key and grade tables published (TCT-DWG-001, this note); fill recipe waits for TRL 4	Published under CERN-OHL-S-2.0	Not verifiable at TRL 3 (partly met)
R1	Standard grades	RT 5 HC 5 to 6 °C, RT 25 HC 22 to 26 °C, RT 70 HC 69 to 71 °C (datasheets)	C5 4 to 6, C25 22 to 27, H70 68 to 72 °C	Met
R3	Stored energy	C5 65.4, C25 60.5, H70 72.9 Wh [B1]	60 Wh or more	Met (C25 thin)
R4	One-hand carry	2.90 to 2.98 kg [A6]	3.0 kg or less	Met (thin)
R5	Efficient use of mass	38.1 to 39.6 % PCM [A6]	38 % or more (stock tube); 50 % for a later extrusion	Met (thin)
R8	No pressure hazard	0.00 bar at fill temperature; +0.29 bar 20 K over the limit; walls 56 MPa under vacuum [C1 to C3]	0.5 bar or less, 10 % ullage	Met
R11	Hot-holding time	13.6 h with the dark finish (7.2 h bare) [D6]	4 h at 63 °C or more	Met
R14	Safe to handle when charged	Bail 39.1 °C on pad charge; 85 °C for about 10 min after an oven, covered by the 12 min label rule [F1, F2]	Below the 10 s burn threshold, with the H70 oven rule	Met with the handling rule
R2	Common envelope	323.4 x 152.4 x 63.5 mm [A1]	GN 1/3, 65 mm	Met
R9	Wrong grade cannot fit	All pairings rejected [G1]	Keyed	Met by design
R13	State visible	Clear melt indicator tube holding the same PCM (TCT-DDR-003, P9)	Readable at a glance	Met by design
R15	Compatible materials	Aluminium, FKM, epoxy rated 120 °C or more, paraffin only	No corrosion or swelling	Met by design (paraffins)

ID	Requirement	Value	Target	Status
R18	Recoverable	G 1/2 port drains the melt; fill named on label	Drain and reuse	Met by design

Counts: 1 not met (R10), 1 over the value-engineering target (R16), 1 at risk, 3 not verifiable at TRL 3, 12 met (8 by calculation or with a rule, 4 by design). In v0.2 R16 was counted as not met against the budget; the change is one of wording (STANDARDS section 18), and the cost rose by \$4 with the constructable design.

■ Checks against the TRL 2 figures

Table 6. TRL 2 claims checked.

TRL 2 claim (TCT-PRC-001 v0.2)	This note	Action
322 x 152 x 61 mm	323.4 x 152.4 x 63.5 mm with the folding bail and stock 6.35 mm fin bar	Precis and README updated
1.62 L, about 1.1 kg PCM	1.655 L; 1.10 to 1.18 kg	Updated
C5 55, C25 61, H70 67 Wh	65.4, 60.5, 72.9 Wh	Updated; R3 now met for C5
About 2.6 kg filled, 42 % PCM	2.90 to 2.98 kg, 38 to 40 %	Updated; R4 thin; R5 restated to 38 % by TCT-DDR-002
Two C5 hold a 25 L cooler for about 8.1 h	Energy lasts 9.7 h, but air-side coupling limits the hold to 1.9 to 2.4 h	R10 changed from at risk to not met
One H70 holds a GN carrier for about 6 h	7.2 h bare, 13.6 h with the decided dark finish	Updated
C5 recharge 4 to 8 h in a freezer; H70 2.5 h on a pad	5.6 h; 2.7 to 10.3 h	Updated; pad at risk
Charge H70 at 100 °C or less	RT 70 HC limit is 90 °C	R12 target and safety text corrected
About \$49 per cartridge, \$15 per frame	\$76 (with the dark finish) and \$16	R16 changed from met to not met